

Heart Rate Variability: A Methodological Survey

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Abstract—Heart rate variability (HRV) is said to be a promising and important investigation tool for a diagnosis of cardiovascular diseases. The regulation and control of the HRV are carried by the parasympathetic nervous system (PNS) and sympathetic nervous system (SNS) of the autonomic control system (ANS). Moreover, the various cardiological and non-cardiological diseases can be investigated from the HRV analysis. Therefore, numerous applications have been presented by the researches using the HRV analysis. In this paper, we are presenting the methodological evaluation in HRV analysis and their findings.

Keywords- Heart rate variability, linear measures, non-linear measures disorders, HRV norms, methodological evolution.

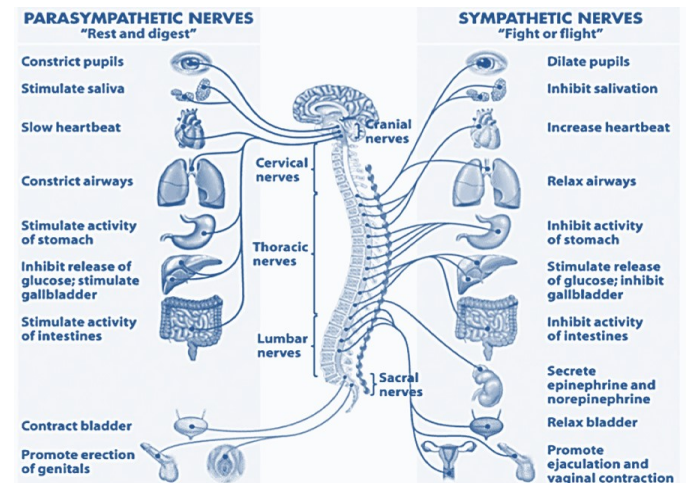
I. INTRODUCTION

The heart is one of the vital organs in the human body which pumps the blood around the body in a particular rhythm [1]. This rhythmic action of the heart is measured using the heart rate (HR). Likewise, the other organs of the human body, diseases, and age of the human affects the performance of the heart which reflects in its rhythmic action [2]. The physiological phenomenon of variation in the time interval between heartbeats (rhythms) is known as Heart rate variability (HRV) [3] which is also known as the variation in the beat-to-beat interval [4, 5].

Basically, the regulation mechanism of HRV is known as a neuro-cardiac function [5]. This heart-brain interaction is controlled by the autonomic nervous system (ANS) as shown in figure 1. As HRV analysis involves ECG signal, it replicates the adaptive ability of the heart for the unpredictable changing circumstances by analyzing and responding to those stimuli [6]. Therefore, HRV is said to be an imperative investigation tool for detecting the functioning of the sympathetic and parasympathetic systems of the ANS for regulating cardiac activity [1-5].

The branches of ANS i.e. SNS and PNS are responsible for the regulation and control of the various functions such as contraction and dilation of pupil, stimulation and inhibit saliva, increase and decrease of the heart rate, respiratory rate, etc. [7] (see figure 1). Any abnormality of the autonomic activity reflected in HRV is associated with the various cardiologic disorders such as blood pressure (BP) [8-12], myocardial infarction, cardiac arrhythmia, Diabetes mellitus [13-17], obesity [18-19] etc., and non-cardiologic disorders [20-24] such as drowsiness, stress, [20], renal failure etc. The details of the HRV application are presented in [5] [25-26]. HRV analysis becomes the most popular and promising cardiac biomarker to investigate the abnormalities associated

with bodily functions. Therefore, different HRV applications and methodologies such as linear [27-33] and non-linear [34-39] are investigated by the various researchers for numerous physiological and clinical findings.



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Fig. 1. Regulation mechanism of the ANS for various bodily functions.

The HRV measure has a common procedure to follow which includes: ECG data acquisition, preprocessing, R-peak detection, R-R interval, HRV methods (linear or non-linear) and are given in figure 2. Before calculating the HRV, accurate detection of the R-peak gains significant importance as it is a key component in most of the methodologies for HRV analysis.

Mainly, a methodology for HRV analysis has two categories viz; linear method and non-linear method. The linear method is the most commonly used technique which includes the time domain and frequency domain analysis of the acquired ECG signal. The power spectral analysis of high frequency (0.15-0.40 Hz), low frequency (0.04-0.015 Hz) and very low frequency (0.003-0.04 Hz) are most frequently used as it reflects sinus rhythm, baroreceptors control, and regulation of vascular and thermal respectively. [1]. On the other hand, the non-linear methods are also been adopted by the researchers to investigate the HRV in recent past [34-39]. Eventually, analysis of HRV requires to follow standard norms. These norms are; 1) long-term (LT) HRV analysis also called as 24h and is most useful for the investigations of the various physical (like myocardial infarction) [40] and psychological (like

Table I: HRV measures [5]

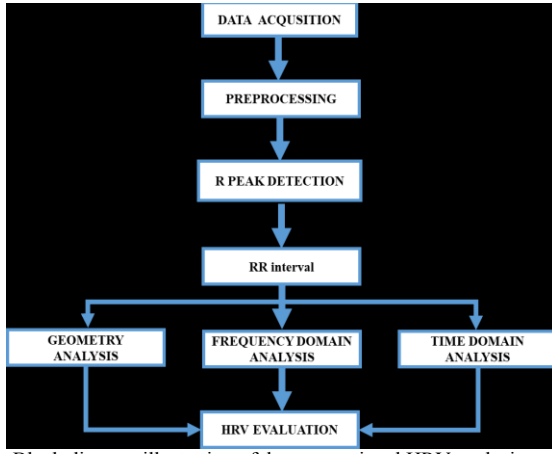


Fig. 2: Block diagram illustration of the conventional HRV analysis approach.

depression) [41] conditions. As this is an ambulatory monitoring, it suffers from the problems of high non-stationary and low reproducibility [42-45]. 2) Further, to overcome the problems of long-term analysis, new analysis criteria had been proposed in which a recording of a short duration of 5 min is used for HRV analysis and is called as Short-term (ST) HRV analysis (5 min). It also holds one more advantage that it is less dependent on the recording conditions which makes it suitable for the monitoring a short-term ANS response to stimuli [39]. 3) However, 5 min of recording is also found unsuitable for real-time applications like e-health and telemedicine [39]. This leads to an increase in the demand of the ultra-short term (UST) analysis (less than 5 min). This quick investigation of HRV using UST has several prognostic significances for patients with cardiac risk and gaining significant demand in the medical domain [39].

This paper presents the methodological evaluation commonly and recently been used since 2014 in HRV analysis. A very recent article on the same was published in 2014 [25], however, there are numerous HRV methodologies and applications have been published thereafter. Therefore, in this paper, we are mainly focusing on those articles which are published after 2014.

The paper is organized as: the details of linear methods are presented in section II. Section III deals with the non-linear methods used for HRV analysis. The performed experiments and its results using the linear and non-linear HRV analysis on MIT-BIH Arrhythmia database from PhysioNet. Finally, the general findings are presented to conclude the paper.

II. LINEAR METHODS OF HRV

since late '20s, there are several studies which involve HRV analysis using linear methods alone [27-33] due to its ease of implementation. It consists of two domains: 1) Time domain and 2) frequency domain which is explained as follows:

A. Time-domain measures

Time-domain measures further divided into statistical measures such as SDNN, SDRR, SDANN, SDNN index (SDNNI), pNN50, HR Max – HR Min, RMSSD, and geometric measures like HRV triangular index (HTI), TINN, Differential index, and Logarithmic index. The time-domain

Metrics	Parameters	units	Description
Time-domain	SDNN	ms	The standard deviation of NN intervals
	SDRR	ms	Standard deviation of RR intervals
	SDANN	ms	The standard deviation of the average NN intervals for each 5 min segment of a 24 h HRV recording
	SDNN index (SDNNI)	ms	Mean of the standard deviations of all the NN intervals for each 5 min segment of a 24 h HRV recording
	pNN50	%	Percentage of the difference greater than 50ms between successive RR intervals
	HR Max – HR Min	bpm	Average difference between the higher and lower HR during each cycle
	RMSSD	ms	Differences between root mean square of successive RR intervals
	HRV triangular index (HTI)		The ratio of integral of the density of the RR interval histogram and its height.
	TINN	ms	Baseline width of the RR interval histogram
Frequency-domain	ULF power	ms ²	Absolute power of the ultra-low-frequency band (≤ 0.003 Hz).
	VLF power	ms ²	Absolute power of the very-low-frequency band (0.0033–0.04 Hz)
	LF peak	Hz	Peak frequency of the low-frequency band (0.04–0.15 Hz)
	LF power	ms ²	Absolute power of the low-frequency band (0.04–0.15 Hz)
	LF power	nu	Relative power of the low-frequency band (0.04–0.15 Hz) in normal units
	LF power	%	Relative power of the low-frequency band (0.04–0.15 Hz)
	HF peak	Hz	Peak frequency of the high-frequency band (0.15–0.4 Hz)
	HF power	ms ²	Absolute power of the high-frequency band (0.15–0.4 Hz)
	HF power	nu	Relative power of the high-frequency band (0.15–0.4 Hz) in normal units
	HF power	%	Relative power of the high-frequency band (0.15–0.4 Hz)
	LF/HF	%	Ratio of LF-to-HF power
Non-linear measures	S	ms	Area of the ellipse which represents total HRV
	SD ¹	ms	Poincaré plot standard deviation perpendicular the line of identity
	SD ²	ms	Poincaré plot standard deviation along the line of identity
	SD ¹ /SD ²	%	Ratio of SD ¹ -to-SD ²
	ApEn		Approximate entropy
	SampEn		Sample entropy
	DFA α_1		Detrended fluctuation analysis for short-term fluctuations
	DFA α_2		Detrended fluctuation analysis for long-term fluctuation
	CD		Correlation dimension

measures are said to be simplest ones which can be calculated right from the RR intervals of a time series data.

The detailed description of time-domain parameters is given in table I. The mathematical representations of the time-domain measures are as follows:

$$SDNN = \sqrt{\frac{\sum_{i=1}^N (RR_i - \overline{RR})^2}{N-1}} \quad (1)$$

$$RMSSD = \sqrt{\frac{\sum_{i=1}^N (RR_{i+1} - RR_i)^2}{N-1}} \quad (2)$$

$$NN50 = (RR_{i+1} - RR_i) > 0.050 \quad (3)$$

$$pNN50 = \frac{NN50}{total\ NN} \times 100\% \quad (4)$$

$$SDANN = \sqrt{\frac{\sum_{i=1}^N RR_i - \overline{RR}}{N-1}} \quad (5)$$

$$HTI = \frac{c}{d} \quad (6)$$

where, RR_i - i^{th} RR interval, $\overline{RR}$ - average RR interval, c - the total number of the RR interval histogram and d - the height of the RR interval. Most of the studies have considered only time domain measures, further, the noticeable properties of parametric and non-parametric spectrum estimation have made frequency domain measures popular for HRV analysis [25].

B. Frequency-domain measures

Frequency-domain measures consist of ULF power, VLF power, LF peak, LF power, HF peak, HF power, LF/HF which are explained in table I. In case of the frequency-domain measures, it is required to evenly sample the time series to analyze its spectral information uniformly. Therefore, the samples are interpolated before its spectral analysis. To further estimate the spectral analysis, either parametric based or non-parametric based method is used to calculate the power spectral density (PSD) of RR intervals. In this process, various frequency bands such as very low frequency (VLF, 0-0.04 Hz), low frequency (LF, 0.04-0.15 Hz), and high frequency (HF, 0.15-0.4 Hz) are used to calculate the power and peak frequency to analyze PSD. The details of the existing linear methods for HRV analysis and its applications are presented in Table II.

III. NON-LINEAR METHODS OF HRV

Even though the linear methods are easier and most commonly used till last decades for HRV analysis still it is not sufficient to characterize the intricate dynamics mechanisms involved in cardiovascular regulation. Therefore, researchers are considering non-linear measures together with linear measures to provide a precise investigation of the HRV [46-55]. The most conventional non-linear parameters are Poincare' plot analysis (S , SD^1 , SD^2), Approximate entropy (ApEn), Sample entropy (SampEntropy), Detrended fluctuation analysis (α_1 , α_2), Shannon entropy (ShanEntropy), and Multiscale entropy (MSE). Although, many parameters are introducing and developing day by day still very few have managed to established clinical utility [39]. The detailed description of the non-linear measures is as follows:

A. Poincare' plot analysis (S , SD^1 , SD^2)

Poincare' plot is a quantitative analysis performed by fitting an ellipse to the generated scatterplot of the successive RR intervals (RR_m and RR_{m+1}). From this plot, several parameters are extracted for HRV investigation among them S , SD^1 , SD^2 is the most commonly calculated due to its ease of implementation [39]. The details of S , SD^1 , SD^2 are given in table I and are mathematically expressed as:

$$SD^1 = \frac{1}{2} (SDSD)^2 \quad (7)$$

$$SD^2 = 2 (SDNN)^2 - \frac{1}{2} (SDSD)^2 \quad (8)$$

where SDNN is Standard deviation of all NN intervals, and SDSD is Standard deviation of differences between adjacent NN intervals. Recently, authors have also proposed a lagged Poincare' plot method for the reliability study, where scatterplots are between RR_m and RR_{m+M} , M is lag ranges from 1-10 [39].

B. Approximate entropy (ApEn)

In 1991, a non-linear measure called ApEn which quantifies the complexity, unpredictability, and irregularity or randomness of the RR interval time. ApEn is used to quantify the probability of the self-similarity or repetitive patterns which can be indicated as a positive number, a small value close to zero reflects higher similarity while the high positive value of ApEn indicates less repetitive patterns in the RR time series (more complexity, and irregularity). ApEn has two main limitations: 1) ApEn is extremely reliant on the length of the recording, hence, it is evenly poorer for short records. 2) ApEn lacks comparative consistency [25]. Therefore, it becomes a complex task for the comparisons cases as ApEn of one data set is higher than that of another then it may not remain higher for all conditions tested. ApEn is associated with the vagal modulation and is high correlates with HF power time series [25].

$$ApEn(S_N, m, r) = \ln \left[\frac{C_m(r)}{C_{m+1}(r)} \right] \quad (9)$$

$$C_{im}(r) = \frac{n_{im}(r)}{N - m + 1} \quad (10)$$

where $n_{im}(r)$ is the number of self-similar patterns in p_m and $C_{im}(r)$ is the segment of self-similar patterns of length m at interval i . S_N is ' N ' instantaneous heart rate measurements $\{HR(1), HR(2), \dots, HR(N)\}$

C. Sample Entropy (SampEntropy)

Further, to overcome the limitation of ApEn i.e. highly data size dependency and inconsistency, a new measure which relates time series regularity and is called as sample entropy (SampEn) [25]. The main advantage of the SampEn is it finds relatively consistent in the situation where the ApEn was failing to do so [25]. A lower value of SampEn is reflecting regular time series while high value shows random time series. Many researchers have found improvement in the accuracy on adopting SampEn which ensures its usefulness in clinical and biological findings with time series data.

Table II: Methodological review

S. N.	Authors	Linear/Nonlinear	Application	Findings
1.	Munoz et al., 2015 [27]	Linear	Feasibility analysis of ultra-short-term measurement	10 s recording of ECG is sufficient to analysis the HRV using time domain methods.
2.	Aydin, S. G et al., 2016 [28]	Linear	Congestive Heart Failure	A significant difference between HRV measures for the normal and congestive heart failure is observed
3.	Lahdenoja, O et al., 2016 [29]	Linear	HRV analysis	Combination of the axes of accelerometer and gyroscope provided more reliable HRV.
4.	Tobón, D. P et al., 2017 [30]	Linear	HR and HRV analysis from noisy signal	The proposed methods outperform for analysis of the HR and HRV from the noisy signals with comparable less computation time.
5.	Siecinski, S et al., 2017 [31]	Linear	HRV analysis for rumination and depressive symptoms	Resting HRV does not have any correlation between the rumination and depression while in case of ambulatory setting, low vagal tone may be a cause for the generation of depressive symptoms
6.	Carnevali, L et al., 2018 [32]	Linear	HRV Analysis from SCG and ECG	HRV from the Seismocardiography (SCG) is highly correlated with the HRV from ECG signals.
7.	Negulyaev, V. O et al., 2018 [33]	Linear	HRV analysis from ECG and PPG	Pulse interval and RR intervals are coinciding in low frequency region but found significant difference in high frequency region.
8.	Gospodinova, E et al., 2015 [34]	Non-linear	HRV analysis using graphic-oriented	The information of the physiological condition of the patient can be analyzed using the proposed method and also, the graphic-orientation analysis is helpful for the medical expert to analyze the large number of data.
9.	Nardelli, M et al., 2015 [35]	Nonlinear	gender-specific classification system	Non-linear dynamics found to be affected by Caress-like stimuli elicitation withgender dependency
10.	Martin, A et al., 2015 [36]	Nonlinear	EEG and HRV signals during sleep	Fluctuation in HR and Delta band energy have shown noticable correlation.
11.	Wendt, H et al., 2018 [37]	Nonlinear	Risk stratification in Congestion Heart Failure (CHF)	Non-Gaussian multiscale representations provided robust information for risk stratification in CHF subjects.
12.	Mengyao, H et al., 2018 [38]	Nonlinear	Heart Rate Data Based on Chaos Theory	These exponents provide new indicators of cardiovascular diseases and new methods for clinical diagnosis and treatments for some heart diseases
13.	Nardelli, M et al., 2018 [39]	Nonlinear	Reliability of ultra-short-term measurement	The performance of the 35s of data using lagged Poincare' plot parameters is sufficient to achieve correlation with 1 h recording with less error.
14.	Poddar, M. G et al., 2015 [45]	Both linear and non-linear	HRV analysis of healthy subjects of different age	PCA-PNN classifier module classifies better with different age group
15.	Munla, N et al., 2015 [46]	Both linear and non-linear	Driver stress level detection	Stress detection proved good accuracy using SVM-RBF classifier robustness of ECG biometric as an accurate physiological indicator of a driver state
16.	Negru, R. D et al., 2015 [47]	Both linear and non-linear	HRV analysis	Linear parameters found best suitable for investigation of autonomic dysfunction for alcoholic etiology while investigation of hepatitis C can be done more significantly using fractal based non-linear parameters.
17.	Castaldo, R et al., 2016 [48]	Both linear and non-linear	Detection of mental stress	Machine-learning approaches such asMultilayer Perceptron (MLP), Naïve Bayes (NB), AdaboostM1 (AB), decision tree using C4.5, Support Vector Machine (SVM), were used to discriminate the stress and non-stress classes based on HRV features. Among them, decision tree using C4.5 was provided higher performance.
18.	Perpiñan, G et al., 2016 [49]	Both linear and non-linear	Classification of Metabolic Syndrome Subjects and Marathon Runners	Long-term and short-term HRV parameters with KNN classifier shown better performance than conventional parameters like complexity and regularity of the HR.
19.	Severeyn, E et al., 2016 [50]	Both linear and non-linear	The early detection of dehydration using the ECG signal in Athletes	The significant value post statistical analysis found to be decreasing between the rest to post exercise state, while it is again increased from post exercise to post hydration. This will help to earlier detection of the dehydration in athletes.
20.	Narin, A et al., 2017 [51]	Both linear and non-linear	Paroxysmal Atrial Fibrillation (PAF) attack detection using HRV measurements	Among linear and non-linear based PAF attack detection, linear measurement using KNN provides 82% of accuracy. Moreover, it is able to predicts 12.5 mins earlier than the actual attack.
21.	Oh, J, et al., 2017 [52]	Both linear and non-linear	HRV analysis in patient with Delirium in ICU	More irregularities in the HR was found in the patient with Delirium than non-delirium patients.

The sample entropy can be defined mathematically as follows:

$$\text{SampEn}(m, r, N) = -\log_B \frac{A}{B} \quad (11)$$

where, m is the embedding dimension, r is tolerance and N is the data, A= number of template vector pairs having $d[X_{m+1}(i), X_{m+1}(j)] < r$ of length (m+1), and B= number of template vector pairs having $d[X_m(i), X_m(j)] < r$ of length (m).

D. Detrended fluctuation analysis (α_1 , α_2)

In 1995 another non-linear analysis was developed called as detrended fluctuation analysis (DFA) to study the fractal correlation properties or occurrences of intrinsic self-similarity in time series data [25]. Generally, FDA or index α consists of two components: 1) short-term (α_1), consisting correlations from 4-16 beats which correspond to the vagal modulation, and sympathetic, and 2) long-term (α_2), consisting correlations

from 16-64 heartbeats which only corresponds to sympathetic [25].

The value of the α coefficient decides whether the RR intervals are correlated or not. If the coefficient ' α ' is close to 0.5 then there is no correlation between RR interval and has significant randomness in the dynamics in the time series data. Unlikely, if the value is close to 1.0 then it possesses both high correlations with randomness which shows fractal heart dynamics. $\alpha=1.5$ indicates strong correlation [25].

It has been found from the studies that only 256 intervals of data i.e. ~ 3.5 min of RR recording is sufficient enough to the acceptable estimation of the alpha (α) coefficient. Hence, it is expected that 10 min of RR data is adequate to measure the scaling exponents. However, the parasympathetic and sympathetic controls may influence the alpha value and fail to distinguish between them completely. Therefore, it is required a separate analysis of both short-term scale and long-term to reveal the exact range of the scale as it cancels the mutual effect.

E. Correlation Dimension (CD)

The statistics of the available independent components and their nonlinear coupling can be estimated by using the Correlation Dimension (CD). Mainly, the CD is a non-stationary measure which can be evaluated using short series. This overcomes the limitations of the requirement of large time series in other measures [6]. CD is also used to identify the self-similarity patterns. The value of the CD will be more when there is sharp fluctuation in the RR intervals and it reduces during the autonomic balancing by sympathetic controls. The CD is higher in adolescent ranging from 8-17 years while it is found to be decreasing from the age of 22-35 years as the sympathetic activity increases in this age group [25].

IV. EXPERIMENTS AND RESULTS

In this section, we have investigated the HRV using linear metrics such as avg. HRV, RMSSD, SDRR, pNN50, VFL_pLF, pHF, LF/HF and non-linear Poincare plot analysis (SD₁, SD₂ and SD₁₂) which are explained in earlier sections. The experiments are performed on the MIT-BIH arrhythmia database [53] available online. Refer [54] for detailed description of the database. The results of the HRV analysis using the mentioned measures are given the table III.

V. CONCLUSION

In this paper, we have presented the methodological evolution in HRV analysis since 2014. It has been found that although several studies have used time-domain measures alone for HRV analysis still the noticeable properties of parametric and non-parametric spectrum estimation have made frequency domain measures popular for HRV analysis. Certainly, the linear measures have gain importance in HRV analysis due their ease of implementation. However, the non-linear nature of ECG makes it insufficient to characterize the intricate dynamics mechanisms involved in cardiovascular regulation. Therefore, non-linear measures have gained prime

importance for the HRV analysis irrespective of the application. It has also found that there are numerous applications who have utilized the combination of the linear and non-linear measures for the HRV analysis for bettering the investigation.

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Table III: Results of the HRV analysis using linear and non-linear measures

ECG Arrhythmia Database from Physionet [53]											
HRV analysis											
Time-Domain Measurements				Frequency-Domain Measurements					Non-linear Measurements		
Signals	Avg_HRV	RMSSD	SDRR	pNN50	VFL	pLF	pHF	LF/HF	SD ₁	SD ₂	SD ₁₂
100m	1.2876	1.2883	0.0452	30.0000	2.5753	70.8136	29.1864	2.4263	0.0319	0.0533	0.5992
101m	1.1844	1.1849	0.0362	9.0909	2.3680	69.7274	30.2726	2.3033	0.0287	0.0420	0.6850
102m	1.2328	1.2334	0.0415	30.0000	2.4668	38.1115	61.8885	0.6158	0.0324	0.0483	0.6714
103m	1.0017	1.0760	0.4076	42.8571	2.6698	32.3762	67.6238	0.4788	0.3584	0.3527	1.0164
104m	1.0357	1.0361	0.0311	0.0000	2.0688	68.9252	31.0748	2.2180	0.0207	0.0387	0.5345
106m	0.2568	0.3905	0.2972	62.0000	1.2711	27.1457	72.8543	0.3726	0.2856	0.3054	0.9352
108m	0.8645	1.0066	0.5338	60.0000	2.2042	18.2025	81.7975	0.2225	0.5638	0.4220	1.3361
109m	1.3689	1.4193	0.3975	33.3333	2.5876	19.6072	80.3928	0.2439	0.2664	0.2950	0.9030
111m	1.4458	1.5963	0.7177	66.6667	3.2304	26.6987	73.3013	0.3642	0.8084	0.5405	1.4957
114m	1.4418	1.4449	0.1014	66.6667	2.9050	71.6247	28.3753	2.5242	0.1057	0.0917	1.1534
115m	1.0812	1.0813	0.0153	0.0000	2.1628	69.7702	30.2298	2.3080	0.0143	0.0153	0.9295
119m	1.4293	1.4297	0.0359	11.1111	2.8641	72.6082	27.3918	2.6507	0.0245	0.0389	0.6296
121m	1.6205	1.6213	0.0559	50.0000	3.2433	73.8351	26.1649	2.8219	0.0418	0.0581	0.7197
202m	0.9615	0.9616	0.0149	0.0000	1.9252	72.5304	27.4696	2.6404	0.0143	0.0150	0.9494
207m	0.9143	0.9146	0.0265	0.0000	1.8297	69.3393	30.6607	2.2615	0.0157	0.0330	0.4743
210m	0.9731	0.9919	0.1990	71.4286	2.0250	65.3778	34.6222	1.8883	0.2151	0.1692	1.2707
212m	0.9526	0.9559	0.0829	50.0000	1.9078	67.4225	32.5775	2.0696	0.0522	0.0992	0.5263
213m	0.7795	0.7811	0.0508	17.6471	1.5807	69.4609	30.5391	2.2745	0.0545	0.0268	2.0330
214m	0.6612	0.8503	0.5485	55.0000	2.0396	50.1910	49.8090	1.0077	0.2268	0.7174	0.3161
215m	1.4556	1.7765	1.0802	66.6667	4.0604	56.9876	43.0124	1.3249	1.3093	0.7006	1.8690
217m	0.3515	0.5146	0.3809	65.7895	3.9162	30.1406	69.8594	0.4314	0.3070	0.4169	0.7363
219m	1.0520	1.0621	0.1528	75.0000	2.0937	8.7556	91.2444	0.0960	0.1673	0.1127	1.4848
220m	1.2047	1.2048	0.0134	0.0000	2.4107	69.1568	30.8432	2.2422	0.0141	0.0099	1.4223
222m	1.1275	1.1296	0.0742	50.0000	2.2873	51.0815	48.9185	1.0442	0.0815	0.0594	1.3717
223m	1.0800	1.0803	0.0278	8.3333	2.1597	68.7168	31.2832	2.1966	0.0200	0.0330	0.6077
230m	1.0545	1.0547	0.0221	0.0000	2.1058	69.7692	30.2308	2.3079	0.0158	0.0226	0.7018
231m	1.4400	1.4897	0.4048	77.7778	3.0588	66.0759	33.9241	1.9478	0.4319	0.3729	1.1580
233m	0.5589	0.6704	0.3779	76.0000	1.7303	58.5651	41.4349	1.4134	0.2515	0.4500	0.5588